The Uncodified Margin

Human Relational Intelligence as a Structural Complement to Algorithmic Systems in Five Emerging Markets

A Field-Evidence Framework with a 2026 Retrospective Assessment

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Note on prior version. This paper substantially extends and supersedes Gupta, L. (2025), *Bridging the Gap: Blending Human Insights and Digital Tools to Master Emerging Markets*, SSRN Working Paper 5140987. The present version restates the original field cases with the sourcing and case-study method made explicit, adds a formal literature grounding, adds a retrospective assessment of the 2025 argument against the empirical record published since, and adds an applied section on the design consequences of the framework. Readers seeking the original February 2025 formulation should cite the earlier working paper.

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Abstract

Algorithmic decision systems can act only on information that has been codified. This paper argues that in emerging-market commerce a structurally significant share of decision-relevant information is never codified at all, and it names that residue the uncodified margin. The margin is not a temporary data-coverage gap that better infrastructure will close. It is an equilibrium outcome in informal and semi-formal economies: counterparties operating outside formal credit, tax, and contracting regimes have positive incentives against creating a durable record, and information carried through trust networks derives part of its value from restricted circulation. Three properties follow. The margin is decision-specific rather than market-specific; it persists under improving data coverage wherever non-codification is incentive-driven; and it sets the ceiling on how much autonomy any system can responsibly be granted over a decision.

The argument is developed from eighteen years of direct field operation across five emerging markets, namely Gujarat (India), Ethiopia, Vietnam, Cambodia, and Uganda, in petrochemicals, FMCG, and agricultural trading. Four dated cases document formal data systems that were accurate about what they measured and simultaneously silent on what determined the commercial outcome: informal trucking substitution during the 2024 Red Sea shipping disruption; a rumour-driven 20 percent collapse in Ethiopian teff prices that propagated through social messaging ahead of any price feed; contractual failure from payment terms denominated in a fiscal calendar the counterparty did not operate on; and counterparty-specific credit judgment during Uganda's mid-2024 liquidity crunch. Each case is stated as a falsifiable proposition rather than an anecdote, and the framework is grounded in the established literatures on tacit knowledge, institutional voids, and economic embeddedness.

The paper then assesses its own February 2025 prediction against the subsequent empirical record. The joint ILO and World Bank background study for the World Development Report 2026 finds generative AI employment exposure of 34 percent in high-income countries against 11 percent in low-income countries across 135 economies, supporting the paper's claim of transformation rather than replacement, and supporting it for the specific structural reason the original advanced. Two risks the 2025 framing did not anticipate are reported alongside: sharply uneven exposure by gender and by career stage, and a developing-economy sequencing risk in which disruption may arrive faster than augmentation gains. A closing section derives design rules for organisations building market-intelligence systems that operate near the margin.

1. Introduction and Research Question

In early 2025, public and corporate discourse on generative AI was dominated by a displacement framing: that large language models and autonomous agents would substitute for human judgment across a widening range of occupations, including functions such as market entry, supplier negotiation, and distribution strategy that are central to international business in emerging markets. This paper, written in that climate, posed a narrower and more falsifiable question than the prevailing debate allowed. Not whether AI would affect employment in aggregate, but specifically: can algorithmic and data-driven decision systems substitute for human relational intelligence in operating environments characterised by informal credit networks, religious and agricultural calendars, rumour-mediated information flow, and trust-based contracting? These are conditions that are the norm rather than the exception across large parts of emerging-market commerce.

The original answer, drawn from direct field experience rather than survey data or modelling, was that the two forms of intelligence are structurally complementary rather than substitutable. Algorithmic systems excel at processing large-scale, codified, quantitative signals such as

shipment manifests, satellite port data, and price indices. Human relational intelligence is required to interpret and act on information that is local, informal, culturally embedded, and frequently uncodified by design, for reasons ranging from regulatory avoidance to the simple absence of digital infrastructure.

1.1 Contribution

This paper makes three contributions. First, it introduces and defines *the uncodified margin*: the share of decision-relevant information in a given market that is not merely un-digitised but structurally resistant to digitisation, because the parties generating it have positive incentives against codifying it or lack any mechanism to do so. The distinction matters because a coverage deficiency shrinks as data infrastructure improves, whereas a structural margin does not, and the two imply opposite investment strategies for any firm building decision systems in these markets.

Second, it supplies four dated, mechanism-level field cases from five emerging markets in which the uncodified margin was commercially decisive, each stated as a falsifiable proposition rather than an anecdote.

Third, it subjects its own 2025 prediction to assessment against the empirical record published since and reports both where the claim held and where it was incomplete. Practitioner frameworks are rarely revisited against subsequent evidence once published; this paper treats that revisiting as an obligation rather than an option, and Section 5.2 states the two respects in which the original framing understated risk.

1.2 Methodological Note on Case Material

The case material in Section 3 is drawn from the author's direct field experience operating across emerging markets over an eighteen-year period in petrochemicals, FMCG, and related trading sectors. To protect the privacy of counterparties who did not consent to be identified, all personal names, exact company identities, and certain specific locations have been withheld or generalised; role and country or region are retained where doing so does not risk identification. Quoted statements are reconstructed from memory and contemporaneous field notes to convey the substance of what was communicated in each instance; they are not verbatim transcripts. Where a single voice is used to represent a recurring pattern observed across multiple similar interactions rather than one identifiable individual, this is marked as an illustrative composite, and the underlying pattern rather than the specific wording is the evidentiary claim being made. This is a practitioner case-study method in the tradition of theory-building from field cases (Eisenhardt, 1989; Yin, 2018), not a quantitative or randomised research design, and its limitations are addressed in Section 8.

2. Literature Context: Tacit Knowledge, Institutional Voids, and the Limits of Codified Data

The claim that local, relational knowledge resists codification and is therefore difficult to substitute with data systems has a long lineage in international business research, predating the current generation of AI tools by decades.

The Uppsala Internationalisation Model (Johanson & Vahlne, 1977) argued that firms expand into unfamiliar markets incrementally, in proportion to experiential knowledge accumulated through direct operation. The model explicitly distinguished that knowledge from objective, transferable market data, on the grounds that experiential knowledge is acquired only through direct

involvement in foreign operations and cannot readily be taught or purchased. The closely related concept of psychic distance, the sum of factors such as language, culture, business practice, and political system that impede the flow of information between a firm and a market, was developed in the same tradition to explain why formally similar markets can behave very differently in practice.

Polanyi's (1966) distinction between tacit and explicit knowledge, captured in the observation that "we can know more than we can tell," underlies that literature and is directly relevant to the cases that follow. Nonaka (1994) developed the organisational consequence: knowledge conversion between tacit and explicit forms is costly, partial, and lossy. An algorithmic system can act only on what has been made explicit and codified into data.

Khanna and Palepu's (1997, 2010) work on institutional voids supplies the emerging-market specificity. Where market intermediaries such as credit bureaus, enforceable contract regimes, reliable logistics infrastructure, and market research providers are absent or unreliable, firms substitute relational and reputational mechanisms for institutional ones. Granovetter (1985) and Uzzi (1997) characterise the general form of that substitution: economic action is embedded in social relations, and embedded ties carry information that arm's-length ties do not.

The informal-economy literature supplies the scale. Hart (1973) established informal income activity as a structural rather than transitional feature of developing economies, and La Porta and Shleifer (2014) document its persistence at high shares of employment and output across low- and middle-income countries. De Soto (2000) frames the corresponding asset problem: economically real holdings that no formal register records.

On the labour-market side, the relevant frame is the task-based account of automation. Autor (2015) and Acemoglu and Restrepo (2019) argue that technology displaces tasks rather than occupations, and that the net employment effect depends on the rate at which new tasks are created alongside. The recent experimental literature on generative AI supports an augmentation reading within that frame: Dell'Acqua et al. (2023), in a field experiment with 758 consultants, found substantial quality and speed gains inside the tool's competence boundary and degraded performance outside it, and Noy and Zhang (2023) found comparable productivity gains concentrated among lower-performing workers on codified writing tasks.

2.1 The Gap This Paper Addresses

Each of these literatures treats non-codification primarily as a cost or friction: knowledge is expensive to convert, institutions are missing, infrastructure is absent. The implicit expectation is that the residue shrinks as digitisation advances.

This paper's departure is to argue that in informal and semi-formal emerging-market commerce, a material share of non-codification is not friction awaiting resolution but an equilibrium outcome. Counterparties operating outside formal tax, credit, and contracting regimes have positive incentives against creating a durable record of their transactions; information transmitted through trust networks derives part of its value from its restricted circulation; and calendars, obligations, and reciprocity structures organised around religious or agricultural cycles have no formal register to be written into. A data system therefore does not converge on this information as coverage improves. That is the specific claim the field cases below are offered to support, and it is the claim Section 5 tests against subsequent evidence.

3. Field Evidence: Where Data Systems Missed What Relationships Caught

Each case below follows a consistent structure: the formal data signal available at the time, the on-the-ground reality that diverged from it, and the generalisable proposition the case supports.

3.1 Supply Chain Rerouting Under Geopolitical Disruption (Gujarat, India)

Context: Houthi attacks in the Red Sea and escalating Iran-related tensions disrupted an estimated 10 percent of global oil shipments during 2024. Coverage of the disruption focused almost entirely on benchmark crude pricing.

Data-only view: Real-time satellite and port-status data showed specific shipping lanes and ports as blocked or severely delayed, with no readily available signal on viable alternatives.

Ground-truth correction: Small-scale lubricant manufacturers in Gujarat continued sourcing and distributing product by relying on informal trucking networks and driver relationships that routed around blockaded ports. That logistics knowledge exists only in the operational relationships between manufacturers, transporters, and drivers, and is not captured in any shipping-data feed.

> *Real-time data showed blocked ports, but our survival depended on knowing the truckers who found hidden routes.* (Illustrative composite, lubricant sector, Gujarat)

Proposition 1: Geospatial and shipping data describe the failure of formal infrastructure accurately, and do not describe the informal substitution networks that determine whether a business continues operating through a disruption. Formal-infrastructure telemetry is therefore a valid measure of disruption and an invalid measure of resilience.

3.2 Rumour-Driven Price Collapse in an Informal Commodity Market (Ethiopia)

Context: A 20 percent drop in teff grain prices in an Ethiopian market was initially recorded and explained internally as an ordinary market correction.

Data-only view: Available price and volume data confirmed the drop had occurred but offered no causal explanation consistent with underlying supply or demand fundamentals.

Ground-truth correction: Direct conversation with farmers and local traders revealed the actual cause. An unverified rumour circulating on social messaging platforms about an impending export ban had triggered panic selling. The rumour had no basis in confirmed policy, but it propagated faster through informal social networks than any government or market report could confirm or deny it.

> *A WhatsApp rumour wiped out our margins overnight. Data reports caught up a week later.* (Illustrative composite, grain trading, Ethiopia)

Proposition 2: In markets where informal social messaging functions as the primary information infrastructure, sentiment and rumour propagation are leading indicators, and formal price and volume data register the consequence only after the fact. A system observing only the price series will therefore correctly detect the move and systematically misattribute its cause.

3.3 Contractual Failure from Cultural Miscalibration (Vietnam, Cambodia)

Context: A negotiated supply contract with a Vietnamese rice distributor was terminated unilaterally by the counterparty shortly after signing. Separately, a Cambodian rice mill operation experienced sustained friction over payment terms.

Data-only view: Market entry research had relied on aggregate reporting of Southeast Asia's agricultural growth trends, which did not surface the informal credit relationships underlying actual distributor operations.

Ground-truth correction: The Vietnamese distributor's objection centred on a mismatch between the contract's assumptions and his actual operating relationships with informal lenders, a structure invisible in sector-level growth reporting. In Cambodia, a standard "Net 45 days" payment term was operationally meaningless to a mill owner whose cash flow and obligations were organised around the Buddhist lunar calendar and associated harvest festivals rather than the Gregorian fiscal quarter.

> *Our financial model made perfect sense until it clashed with the harvest cycle.* (Illustrative composite, supply chain management, Cambodia)

Proposition 3: Standardised contractual and financial terms encode calendar and credit assumptions from the contracting firm's home market. Where the counterparty's actual obligations are organised on a different calendar, the terms are not merely suboptimal but semantically void, and the failure will present as counterparty unreliability rather than as a specification error.

3.4 Relationship Capital as Crisis Buffer (Uganda)

Context: A nationwide liquidity crunch in Uganda in mid-2024 caused multiple buying counterparties to shift abruptly from advance payment terms of 20 to 80 percent to demands for 120-day Letters of Credit, materially altering the cash-flow risk of continuing to supply them.

Data-only view: Macro-level liquidity and credit data confirmed the crunch was real and economy-wide but provided no firm-specific basis for deciding which counterparties to extend flexible terms to and which to decline.

Ground-truth correction: A deliberately built network of relationships with distributors and financial institutions, accumulated over time, allowed selective, relationship-informed judgment: extending flexibility to counterparties whose underlying business fundamentals were judged sound despite the liquidity disruption, and declining others. The same relationship network subsequently enabled rapid negotiation of a long-term volume contract when an unrelated Middle East disruption to trade routes created a new opportunity.

> *We went from 20 percent up-front to 120-day LC demands almost overnight. For some, we bent over backward. For others, we had to walk away.* (Illustrative composite, regional sales management, Uganda)

Proposition 4: Macro liquidity data identifies that a sector-wide shock is occurring and does not supply the counterparty-specific discrimination needed to respond to any single relationship within that shock. Where no credit bureau covers the counterparty set, that discrimination has no codified substitute.

4. A Working Framework: The Dual Approach

The cases above motivate an operating framework distinguishing what each form of intelligence is structurally suited to capture.

Dimension	Algorithmic / Data Systems ("GPS")	Human Relational Intelligence ("Guide")
Information type	Codified, quantitative, infrastructure-visible: shipment data, price indices, satellite imagery	Uncodified, relational, locally embedded: informal credit, cultural calendars, trust networks

Update speed	Fast once an event reaches a formal reporting system	Often faster for informal-economy events, but unsystematic and person-dependent
Primary failure mode	Blind to information never digitised or formally reported	Limited in scale; dependent on specific individual relationships and recall
Best-suited tasks	Pattern detection across large volumes; tracking known, measurable variables	Interpreting ambiguous signals; counterparty-specific judgment; crisis-time trust
Failure consequence illustrated	Sections 3.1 and 3.2: missed rerouting options; misattributed rumour-driven price moves	Not applicable. The framework argues for combination, not substitution, in either direction
Scaling property	Marginal cost of an additional observation approaches zero	Marginal cost of an additional relationship is roughly constant and high

Table 1. The Dual Approach: complementary, not substitutable, domains of decision-relevant information.

4.1 The Uncodified Margin, Defined

The framework in Table 1 implies a measurable quantity. For a given market and decision class, the **uncodified margin** is the share of decision-relevant information that is structurally unavailable to any data system, as distinct from information that is merely not yet collected.

Three properties follow from the cases.

First, the margin is *decision-specific rather than market-specific*. In the Uganda case, macro liquidity data was fully adequate for the question "is there a system-wide credit shock" and wholly inadequate for the question "should we extend terms to this counterparty." The same market presents a near-zero margin for one decision and a near-total margin for another.

Second, the margin is *persistent under improving data coverage* where non-codification is incentive-driven. Better price feeds would not have surfaced the Ethiopian export-ban rumour, because the rumour was not a price event; it was a belief propagating through a channel with no reporting surface. Better logistics telemetry would not have surfaced the Gujarat trucking substitution, because the substituting parties had no reason to register it and some reason not to.

Third, the margin *sets the ceiling on autonomous operation*. A system permitted to act without human interpretation can be no more reliable than the share of decision-relevant information it can observe. This yields the operational rule in Section 6: the appropriate degree of system autonomy is inverse to the uncodified margin of the decision, and any system that cannot estimate that margin should not be granted autonomy over the decision.

4.2 An Operating Note on Ethics

The relational-intelligence approach described here depends entirely on trust, and trust is forfeited quickly if treated as a data-extraction opportunity rather than a genuine relationship. One operating principle was followed throughout the engagements described in Section 3: declining methods, such as covertly scraping data from a counterparty's systems without consent, that a competitor in one of these markets was observed to use. Trust, once lost with a local network, is difficult to rebuild, and was treated throughout as a constraint on method rather than as a reputational consideration.

This is not incidental to the framework. If the uncodified margin is accessible only through relationships, then extractive method is self-defeating on its own terms: it converts a renewable information source into a spent one.

5. Retrospective Assessment: February 2025 Against the 2026 Record

The original argument was written in February 2025 against a backdrop of acute, largely undifferentiated public concern that generative AI would displace human labour broadly and quickly. Eighteen months on, with substantial new empirical research published by the International Labour Organization, the World Bank, the International Monetary Fund, and academic groups, it is now possible to assess that argument against evidence rather than against the climate it was written into.

5.1 Where the Framework's Core Claim Held

The dominant finding across the major labour-market studies published in 2025 and 2026 is transformation rather than replacement as the modal outcome of generative AI exposure, which was the central claim of the original paper. The ILO and NASK Global Index (ILO, 2025a; ILO Working Paper 140, 2025b) found that one in four workers globally is in an occupation with some generative AI exposure, but that only 3.3 percent of global employment falls into the highest-exposure category where AI could plausibly automate a substantial share of tasks. The large majority of exposed occupations fall into what the ILO terms the augmentation zone, where AI handles specific tasks while humans continue to manage judgment, complexity, and the interpersonal dimensions of the role.

More specifically relevant to this paper's emerging-market focus, the joint ILO and World Bank background study for the World Development Report 2026, covering 135 countries and roughly two-thirds of global employment, found that generative AI employment exposure correlates strongly with national income level: 34 percent of employment in high-income countries is exposed, against 11 percent in low-income countries, with emerging market economies at approximately 25 percent (ILO & World Bank, 2026).

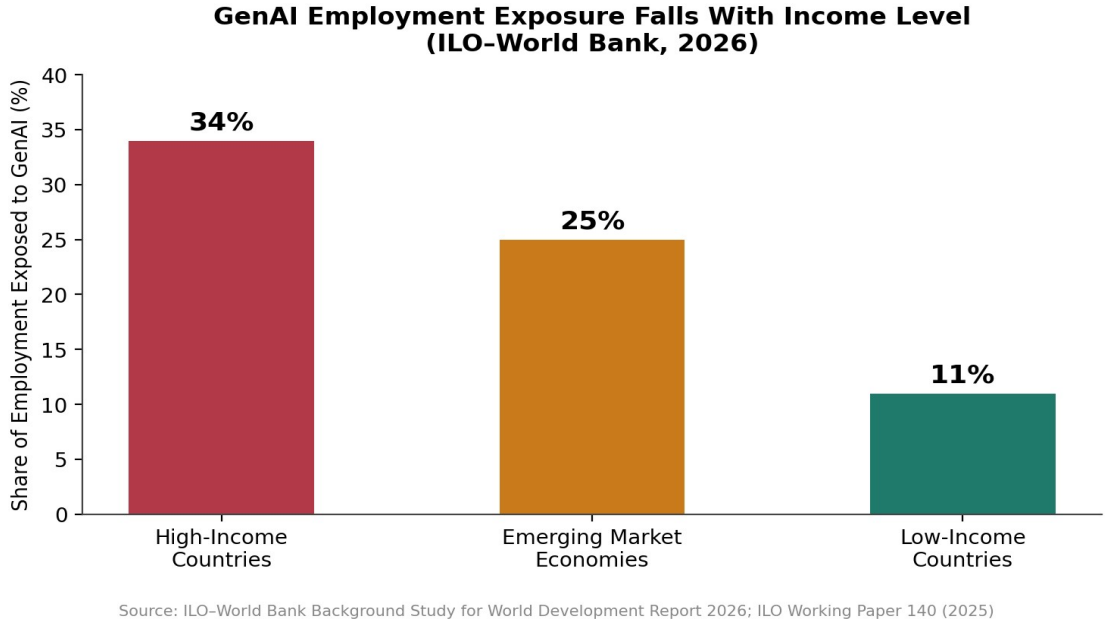


Figure 1. Generative AI employment exposure by income group. The emerging-market context this paper was written about faces structurally lower direct automation exposure than advanced economies. Source: ILO and World Bank Background Study for World Development Report 2026; ILO Working Paper 140 (2025).

This is a more specific and more favourable confirmation of the original thesis than a generic finding that AI augments rather than replaces would be. The study attributes the gap primarily to differences in digital infrastructure and job-task composition. That is the same mechanism the original paper identified qualitatively from the opposite direction: the emerging-market operating environments the field cases were drawn from, characterised by informal credit, low digital-infrastructure penetration, and relationship-mediated information flow, are precisely the environments now independently identified as facing the lowest direct automation exposure, and for the reason this paper argued, which is that a substantial share of decision-relevant information in these markets is not digitised and is therefore inaccessible to an algorithmic system regardless of that system's capability.

The experimental literature reinforces the same pattern in formal-sector settings. Dell'Acqua et al. (2023), in a pre-registered field experiment with 758 Boston Consulting Group consultants, found that consultants using generative AI completed 12.2 percent more tasks and 25.1 percent faster with materially higher quality on tasks inside the model's competence boundary, and performed significantly worse than the unassisted control group on a task designed to sit outside it. That result is structurally identical to this paper's central claim, demonstrated in a formal professional-services setting rather than the informal-economy cases of Section 3: the combination outperforms either component, and the boundary of the tool's competence, not its raw capability, determines where the human contribution becomes load-bearing.

5.2 Where the Original Framing Understated Risk

A fair retrospective must state where the original framing was incomplete. The evidence base published since identifies at least two risk dimensions the original paper did not address, and a balanced revision should not claim a cleaner vindication than the evidence supports.

First, exposure is highly uneven by gender and by career stage in ways the original framework did not model. The ILO's 2025 refined index found the highest-exposure category affects 4.7 percent of female employment globally against 2.4 percent of male employment, with the gap widening in high-income countries to 9.6 percent against 3.5 percent (ILO, 2025b). Separately, Brynjolfsson, Chandar and Chen (2025), using ADP administrative payroll microdata covering millions of United States workers, found that early-career workers aged 22 to 25 in the most AI-exposed occupations experienced a 13 percent relative decline in employment following the post-ChatGPT period, with declines concentrated in occupations where AI substitutes for rather than augments human labour, and with employment stable or growing for more experienced workers in the same occupations. The affected roles are disproportionately those that have historically served as entry points onto a career ladder, a dynamic with direct relevance to the youth-employment concerns raised elsewhere in this research programme (Gupta, 2026c, Section 2). The IMF's own assessment points the same direction from the skills side: Jaumotte et al. (2026) find that new-skill demand is reshaping wages and hiring while deepening polarisation, and that the incidence of new-skill demand in emerging economies is roughly half that of advanced economies, which implies a slower emerging-market on-ramp to the augmentation roles the ILO index identifies as the majority case.

Second, the ILO and World Bank (2026) study identifies a digital-divide risk specific to developing economies that complicates the lower-exposure finding in Section 5.1. Workers in roles vulnerable to automation in developing economies are often already the ones with internet access, even in low-income settings. The jobs most likely to be displaced may therefore displace quickly, while the productivity gains from augmentation are concentrated in roles requiring digital infrastructure that much of the same workforce lacks. The study's authors describe this as a risk that disruption may materialise faster than productivity gains in developing economies, and characterise the associated pattern as a white-collar bypass in which developing economies do not

traverse the clerical and professional employment expansion that advanced economies did. That is a more pessimistic and more conditional finding than the original 2025 framing allowed for.

On balance, the original paper's central, falsifiable claim, that human relational intelligence is a structural complement to rather than a casualty of data-driven systems in the specific informal-economy contexts described, holds up well against subsequent evidence. Its implicit optimism about the pace and evenness of that complementarity across the broader emerging-market workforce, including younger and entry-level workers, was less well supported than the 2026 evidence base now suggests was warranted. The correct reading is that the firm-level case for complementarity is strong and the workforce-level distribution of its benefits is not guaranteed by it.

6. From Framework to Infrastructure: Design Consequences

A framework that identifies a structural information boundary implies design constraints on any system built to operate near it. This section states those constraints and then reports how they were applied in the author's subsequent infrastructure work, offered as a worked application rather than as independent validation of the framework.

6.1 Four Design Rules Derived from Sections 3 and 4

Rule 1: Output evidence and confidence, not directives. If the uncodified margin sets a ceiling on what a system can observe (Section 4.1), a system that emits a directive implicitly asserts that the margin is zero for that decision. A system that emits a structured assessment carrying its reasoning and an explicit confidence value leaves the margin open for a human to close. The difference is architectural, not presentational.

Rule 2: Publish the failures with the successes. Proposition 2 showed a formal data series correctly detecting a move and misattributing its cause. An operator cannot calibrate how much weight to place on a system's output unless the system's error distribution is observable. Reporting only successful cases removes the information the human layer needs to perform its complementary function.

Rule 3: Re-validate against the current regime, not the training regime. Propositions 1 and 4 both turn on abrupt regime changes, a shipping-lane blockade and a national liquidity crunch, in which historical relationships stopped holding. Validation against the period a system was fitted on cannot detect this class of failure.

Rule 4: Do not extract what was given in trust. Section 4.2 states the ethical constraint. Its systems consequence is that information sourced from relationships must not be silently harvested into a data asset, because doing so destroys the source.

6.2 A Worked Application

AION Analytics (India) builds market-intelligence infrastructure for Indian financial markets under a stated public mandate that its outputs inform and do not instruct. The four rules above are visible in that infrastructure as design decisions, and are recorded here because the sequence, field observation to framework to built system, is the substantive claim of this section.

The firm's principal system, AION Indian Market Intelligence, accepts a single macro, policy, weather, or geopolitical event and returns a structured assessment of how that event propagates across Indian market sectors, expressed as sector-level impact together with the causal reasoning, an associated temporal lag, and an explicit confidence value, delivered as structured data rather than as a recommendation. That output shape is Rule 1 implemented: the system reports what it

assessed and how strongly, and the interpretation step remains with the user. It does not issue buy or sell instructions, and is not a signal service.

The firm's pre-open probability model, Veritas, publishes calibrated probability bands for Indian equity indices together with a permanently public validation record that includes sessions in which the band was missed and sessions in which directional conviction was weak. That is Rule 2 implemented. Band widths are adjusted against a trailing window of live sessions rather than against the fitting period, which is Rule 3.

The governing standard the firm publishes for this work, the Enki Protocol, states four commitments: evidence first, with no heuristics; continuous re-validation against the market actually faced rather than the market trained on; output over operator; and privacy treated as an entitlement rather than a configurable setting. The first three correspond to Rules 1 through 3. The fourth is Rule 4 generalised from counterparty relationships to end users.

Two limitations of this section should be stated plainly. First, the correspondence between the framework and the built system is a design correspondence claimed by the author, who is the same person in both roles; it is not an independent test of the framework, and nothing in Section 6.2 should be read as evidence for Sections 3 through 5. Second, the operating mechanisms of these systems are proprietary, and one is the subject of a filed patent application; this section describes design posture and output contracts only, which are matters of public record, and does not describe internal method.

6.3 The Signal-Provenance Gap as a Regulated-Market Instance

The uncodified margin is defined in Section 4.1 for informal markets, but the definition does not depend on informality. It depends only on a mismatch between what a system records and what determines the outcome. That mismatch also appears inside highly formalised, heavily regulated markets, where it takes a specific and consequential form.

In Gupta (2026a), the author documents that India's 2026 regulatory framework for retail algorithmic trading establishes a complete and auditable record of the order chain, from the broker's systems through to the exchange, while the signal chain, meaning what actually caused the order to be generated, remains outside the recorded perimeter. A fully compliant order arrives at the broker with its origin unobserved. The paper evidences that gap from primary regulatory sources in India, the European Union, and the United Arab Emirates, and states it in general form for any framework that anchors accountability at an intermediary while the conduct of interest occurs upstream of that intermediary.

That is the same structure as Proposition 2. The formal system measures accurately what it measures, the measurement is complete with respect to its own perimeter, and the causal origin sits outside the perimeter. Ethiopia's price feed recorded the collapse and not the rumour; a compliant order log records the order and not the signal. The generalisation is that regulatory and data architectures both define a perimeter, and the uncodified margin is whatever determines outcomes from outside it. Formalisation moves the perimeter; it does not by itself eliminate the margin.

7. Implications for Practice

- Treat algorithmic outputs as one structured input into a decision rather than as the decision itself, particularly in market segments where a meaningful share of economic activity is conducted through informal or unbanked channels.

- Estimate the uncodified margin for a decision class before granting a system autonomy over it. Section 4.1 gives the rule: autonomy should be inverse to the margin, and a system that cannot estimate its own margin should not hold the decision.
- Invest deliberately in relationship-building with local distributors, financial counterparties, and community intermediaries as a distinct operating capability with its own budget line, not as a soft adjunct to a primarily data-driven strategy. Note from Table 1 that this capability does not scale on the same cost curve as data infrastructure, and should not be planned as if it does.
- Calibrate standardised contractual and financial terms, including payment windows and fiscal-quarter assumptions, against the counterparty's actual operating calendar, which may be religious, agricultural, or otherwise non-Gregorian. Proposition 3 indicates that failures of this type will be misdiagnosed as counterparty unreliability unless specifically checked for.
- Monitor informal information channels such as community messaging platforms and local trader networks as leading indicators in markets where formal price and volume reporting lags actual sentiment shifts, while treating the ethical constraint in Section 4.2 as binding on how that monitoring is conducted.
- For workforce and hiring strategy specifically, attend to the entry-level and gender-differentiated exposure risk identified in Section 5.2, which the original 2025 framework did not anticipate. Augmentation at the firm level does not guarantee an even distribution of its benefits across the workforce, and the emerging-market on-ramp to augmentation roles appears to be slower than the advanced-economy one (Jaumotte et al., 2026).

8. Limitations

- Case material is drawn from a single practitioner's field experience across a defined set of markets and sectors, namely petrochemicals, FMCG, and agricultural trading. It is illustrative rather than statistically representative, and the framework's generalisability beyond these sectors has not been independently tested.
- Quoted field statements are reconstructed from memory and notes rather than from verbatim recordings, and in some cases represent a composite of a recurring pattern rather than one identifiable individual, as stated in Section 1.2.
- The uncodified margin is defined conceptually in Section 4.1 but is not operationalised as a measurable quantity in this paper. Constructing an estimator for it, for example from the divergence between formally reported and survey-observed transaction volumes in a given market segment, is left to future work and is the most direct route by which the framework's central claim could be tested rather than illustrated.
- The retrospective assessment in Section 5 relies on aggregate, cross-country labour statistics from the ILO, World Bank, and IMF that were not designed to test this paper's narrower claim about informal-economy information asymmetries. The directional consistency reported should not be read as formal statistical confirmation of the 2025 framework. The Brynjolfsson, Chandar and Chen (2025) result cited in Section 5.2 is drawn from United States payroll data and its transfer to emerging-market labour markets is an inference, not a finding.

- Section 6.2 reports a design correspondence asserted by the author about the author's own systems. It is offered as an application of the framework and carries no evidentiary weight for the framework itself.
- No claim is made that the specific named locations, being Gujarat, Ethiopia, Vietnam, Cambodia, and Uganda, are representative of their entire national economies. Each case illustrates a mechanism rather than supporting a national-level conclusion.

9. Conclusion

Written in February 2025 against a climate of acute AI-displacement anxiety, this paper argued for a structural division of labour between algorithmic and human relational intelligence in emerging-market operations, evidenced through field cases in which data-only approaches missed information that local relationships caught. The revision names the underlying quantity, the uncodified margin, and states its three properties: it is decision-specific, it persists under improving data coverage wherever non-codification is incentive-driven, and it sets the ceiling on how much autonomy any system can responsibly be granted over a decision.

Assessed against the most rigorous cross-country evidence available as of mid-2026, principally the ILO and NASK index and the ILO and World Bank background study for the World Development Report 2026, the central claim holds. Transformation rather than replacement is the dominant observed pattern, and the specific informal-economy, low-digital-infrastructure conditions this paper's cases were drawn from are now independently identified as facing structurally lower direct automation exposure than advanced-economy, highly digitised occupations, for the same mechanical reason the original argument gave.

The evidence base has also surfaced real distributional risks that the original framework did not address: to entry-level and to female workers specifically, and to developing economies whose most digitally exposed workers may be displaced faster than augmentation gains materialise. A fair accounting of this paper's predictive record includes both the parts that held and the parts that did not anticipate enough.

The wider implication is that the uncodified margin is not a property of poor countries or immature data infrastructure. It is a property of the relationship between any recording perimeter and the conduct that perimeter fails to reach, which is why the same structure recurs in one of the most heavily regulated market environments in the world, as Section 6.3 sets out. Building systems that are honest about where their perimeter ends is, on this argument, not a concession to present technical limits but a permanent design requirement.

Declarations

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Competing interests. The author is the founder of AION Analytics (India). Section 6.2 describes systems built by that firm and should be read in light of this interest. No part of Sections 3 through 5 depends on those systems, and Section 6.2 is explicitly excluded from the paper's evidentiary basis. The author received no external funding for this research.

Commercial disclosure. AION Analytics (India) is not a signal provider and does not issue buy or sell recommendations. Systems referenced in Section 6.2 produce probabilistic, informational, and educational output. Nothing in this paper constitutes investment advice or a

solicitation. The firm is distinct from Aion Analytics LLC (United States) and AION Labs (Israel).

Data availability. This paper draws on non-public field observation and contemporaneous notes that cannot be released without identifying counterparties who did not consent to identification, as set out in Section 1.2. All secondary statistics cited in Section 5 are drawn from publicly available publications by the ILO, the World Bank, the IMF, and the cited academic sources, and are reproducible from those sources.

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